Negative Numbers

Below zero, the numbers get colder.

Read the thermometer



Above 0 the numbers are positive: 5, 10, 15.

Below 0 they are negative. We write a minus sign:

6 below zero is written −6.

The lower it is, the colder: −10 is colder than −6.

This one shows −6 degrees.

Opposites and distance from 0

The opposite of 4 is −4. | −4 | = 4

Opposites are the same distance from 0, one above and one below.
The bars | | mean: how far from 0? It is never negative.

Your turn!

−3 [>] −9

For teachers

Goal: read, compare and order negative numbers, opposites and absolute value (6.NS.5–7) with a thermometer, not a number line. Say “Which is colder? It is lower on the thermometer.”

Negative Numbers: Practice

Lower on the thermometer means colder.

Part 1 Read the thermometer Write the temperature. Below 0, write the minus sign.

1) 

−4 degrees

2) 

7 degrees

3) 

−12 degrees

4) 

3 degrees

5) 

−9 degrees

6) 

13 degrees

7) 

−2 degrees

8) 

−14 degrees

9) 

11 degrees

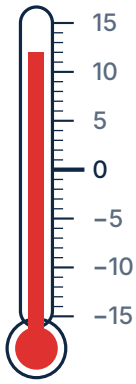
Negative Numbers: Practice (continued)

Part 2 Show the temperature

10) 

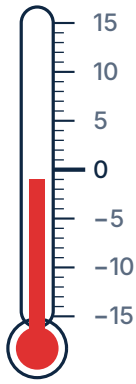
Show −7

Color the tube.

11) 

Show 12

Color the tube.

12) 

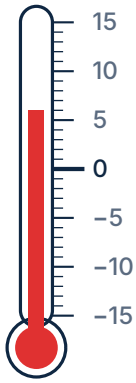
Show −1

Color the tube.

13) 

Show −13

Color the tube.

14) 

Show 6

Color the tube.

15) 

Show −10

Color the tube.

Part 3 Opposites Same distance from 0, the other side.

16) **opposite of 7 =** **−7**

17) **opposite of −3 =** **3**

18) **opposite of 12 =** **−12**

19) **opposite of −18 =** **18**

20) **opposite of 25 =** **−25**

21) **opposite of −9 =** **9**

Negative Numbers: Practice (continued)

Part 3 Opposites (continued)

22) opposite of 41 = −41

23) opposite of −60 = 60

24) opposite of 15 = −15

25) opposite of −1 = 1

26) opposite of 100 = −100

27) opposite of −33 = 33

Part 4 Warmer or colder? Write < or >.

28) −3 > −8

29) −5 < 2

30) −11 < −7

31) 4 > −6

32) −1 > −10

33) −15 < −12

Negative Numbers: Practice (continued)

Part 4 Warmer or colder? (continued)

34) $0 \boxed{>} -4$

35) $-20 \boxed{<} 3$

36) $-9 \boxed{<} -2$

37) $8 \boxed{>} -8$

38) $-14 \boxed{>} -16$

39) $-6 \boxed{<} 0$

40) $-25 \boxed{<} -19$

41) $-7 \boxed{>} -13$

42) $1 \boxed{>} -1$

Part 5 How far from 0? | | means distance from 0.

43) $|-8| = \boxed{8}$

44) $|14| = \boxed{14}$

45) $|-23| = \boxed{23}$

Negative Numbers: Practice (continued)

Part 5 How far from 0? (continued)

46) $|-5| =$ 5

47) $|31| =$ 31

48) $|-50| =$ 50

49) $|-17| =$ 17

50) $|9| =$ 9

51) $|-42| =$ 42

52) $|-3| =$ 3

53) $|66| =$ 66

54) $|-11| =$ 11

Part 6 Coldest to warmest Put them in order.

55) $-2\ 5\ -7\ 0$ in order: −7, −2, 0, 5

56) $3\ -1\ -8\ 6$ in order: −8, −1, 3, 6

57) $-12\ 4\ -5\ -15$ in order: −15, −12, −5, 4

Negative Numbers: Practice (continued)

Part 6 Coldest to warmest (continued)

58) 9 −9 −3 1 in order: −9 −3 1 9

59) −6 −11 2 −4 in order: −11 −6 −4 2

60) 7 −14 0 −10 in order: −14 −10 0 7

Part 7 More warmer or colder Write < or >.

61) −22 < −14

62) −13 > −27

63) −19 < 11

64) −6 > −9

65) 17 > −21

66) −30 < −15

67) −10 < 16

68) −29 < 12

69) −21 < −6

Negative Numbers: Practice (continued)

Part 7 More warmer or colder (continued)

70) $-25 < 19$

71) $3 > -25$

72) $15 > -28$

73) $-12 < 10$

74) $-2 > -26$

75) $-24 > -28$

76) $-26 < 3$

77) $0 > -23$

78) $-16 < -5$

79) $18 > -27$

80) $-20 < 6$

81) $-7 > -24$

Negative Numbers: Practice (continued)

Part 8 More distance from 0

82) $|-77| = 77$

83) $|83| = 83$

84) $|-90| = 90$

85) $|41| = 41$

86) $|-93| = 93$

87) $|-20| = 20$

88) $|-24| = 24$

89) $|-84| = 84$

90) $|-27| = 27$

91) $|-58| = 58$

92) $|-49| = 49$

93) $|42| = 42$

Negative Numbers: Practice (continued)

Part 9 More coldest to warmest

94) 12 −20 −2 −9 in order: −20 −9 −2 12

95) −3 −9 13 2 in order: −9 −3 2 13

96) −7 −14 −4 8 in order: −14 −7 −4 8

97) −1 5 7 −10 in order: −10 −1 5 7

98) 15 −16 −14 10 in order: −16 −14 10 15

99) −11 −7 −14 −12 in order: −14 −12 −11 −7

Part 10 Remember? Problems from earlier packets.

100) GCF of 54 and 90 = 18

101) LCM of 6 and 14 = 42

102) 3.4 × 2.5 = 8.5

103) 25% of 96 = 24

104) 1,782 ÷ 18 = 99

105) $\frac{3}{8} \div \frac{1}{16} = 6$

Negative Numbers: Practice (continued)

Part 11 Word problems Positive or negative?

106) At night it was 8 degrees below zero. Write the temperature as a number.

Number sentence 8 below zero = −8

Answer −8 degrees

Draw or show your work

107) A diver is 30 feet below the sea. A bird is 30 feet above it. Write both heights.

Number sentence −30 and 30

Answer −30 and 30 feet

Draw or show your work

108) Mon. was −5 degrees and Tue. was −9 degrees. Which day was colder?

Number sentence −9 < −5

Answer Tue.

Draw or show your work

109) Sam owes his sister $12. Write it as a negative number.

Number sentence owes $12 = −12

Answer −12 dollars

Draw or show your work

110) Challenge: Town A is at −14 feet and Town B at −22 feet. Which is lower, and by how much?

Number sentence −22 < −14, 22 − 14 = 8

Answer 8 feet, Town B

Draw or show your work

Check-In: Negative Numbers

Show what you know. No help this time.

Mastery: 9 of 11 (80%)

Part 1 Read the thermometer

1) 

5 degrees

2) 

−8 degrees

Part 2 Opposites

3) opposite of 54 = −54

4) opposite of 16 = −16

5) opposite of 6 = −6

Part 3 Warmer or colder? Write < or >.

6) −6 < −1

7) −4 > −9

8) −13 < 2

Part 4 How far from 0?

9) | −45 | = 45

10) | −19 | = 19

11) | 28 | = 28